

# Software Architecture 2024[2026]

## Lecture 3, part 1/3

### Introduction

Dimitri Van Landuyt

Laurens Sion

Wouter Joosen

# Lecture plan

1. Course introduction
2. Quality requirements and architectural significance
- 3. Software architecture: defined**
4. Architectural views
5. Architectural models

...

# This lecture

1. Project Q&A
2. Twin peaks model to SE
3. Software architecture: defined

## › Next lecture

›› Architectural views

# Three complementary strategies

Identify relevant quality attribute requirements at the basis of

1. ...*the functionality of the system*
2. ...*business goals*
3. ...*types of quality attributes*

# Project Q&A

# Some common questions

- › *Should we include  $[L,L]$  ASRs?*

# Some common questions

› *Should we include  $[L,L]$  ASRs?*

1. We are interested in the 15 most relevant ASRs for the design of our system

# Some common questions

› *Should we include [L,L] ASRs?*

2. Therefore, including [L,L] ASRs implies that you are of the opinion that there are no other ASRs left of relevance to our system:

No [L,M], [M,L], [M,M], [H,L], [L,H], [M,H], [H,M], [H,H]



# Some common questions

- › *How detailed should the ASRs be?*

# Some common questions

## › How detailed should the ASRs be?

| Quality Attribute | Attribute Refinement | Summary<br>Rationale <i>Business Value</i><br>Rationale <i>Architectural Impact</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Performance     | Document generation  | <p>The system should be able to process at least <b>60 000</b> documents per hour under normal load (considering 25 customer organizations like Telenet requiring 1.7 million invoices every 30 days).</p> <hr/> <p><b>H:</b> <i>This is very important to meet the time constraints of document generation which is the main service DocProc provides to its customers and specified as part of the SLA negotiated with the customers.</i></p> <hr/> <p><b>M:</b> <i>This requires significant end-to-end performance optimizations at all parts of the document generation process.</i></p> |

- **Precise** in terms of what it applies to (e.g. "the network is disconnected"; "the *user dashboard* does not respond in time")
- Hint towards **Response** and **Reponse measure**

# Some common questions

- › *I'm no expert in  $X$ , how on earth should I come up with a realistic response measure?*

# Some common questions

- › *I'm no expert in  $X$ , how on earth should I come up with a realistic response measure?*
  1. **Expertise and background knowledge helps** – time to brush up basics from other courses and perhaps do some research?

# Some common questions

- › *I'm no expert in X, how on earth should I come up with a realistic response measure?*
- 2. You don't have to be an expert to define an “***interval/threshold of acceptability***”
  - approach this from a customer / business goal perspective (external perspective)

# Some common questions

- › *I'm no expert in X, how on earth should I come up with a realistic response measure?*
- 3. **Realism will not be evaluated** and a lack thereof not reflected in your grades – focus is on correct identification and characterization of the relevant NFRs, the use of appropriate metrics, prioritization

# Some common questions

- › *I'm no expert in X, how on earth should I come up with a realistic response measure?*

## 3. **Realism will not be evaluated**

**Absurdly unrealistic statements** can of course be avoided

- *"We can redevelop the DocProc in 1 Person Month"*
- *"Batch generation should take less than 2 msec"*

# Some common questions

- › *What does the quality attribute scenario add to the ASR summary description?*



# Some common questions

- › *What does the quality attribute scenario add to the ASR summary description?*

## 1. **Scenario aspect:**

**stimulus -> response**

- **Stimulus** is a '*challenge*' to the software quality
- The **response** is how the system *shows its quality (quality attribute response)*

# Some common questions

## 1. Scenario aspect

- **Source:** The marketing department
- **Stimulus:**
  - Wishes to employ DocProc for other types of documents than payslips or invoices.



## Modifiability: New types of documents

- **Response:**

- This modification should not affect the interface for initializing and initiating document processing jobs, except for the functionality related to checking the raw data for errors.
- The interface for registering companies should be extended with the functionality to enable the new types of documents for new customer organizations and configure their template for these types of documents and check these templates for errors.
- The interface for consulting the personal document store and specific documents should be extended with the functionality to show the new types of documents.
- This modification should not affect the systems that generate documents: generating the new types of documents should reuse as much of the existing functionality for generating documents as possible.
- Storing generated documents of the new types of documents should not require changes to the existing system for storing generated documents.

# Some common questions

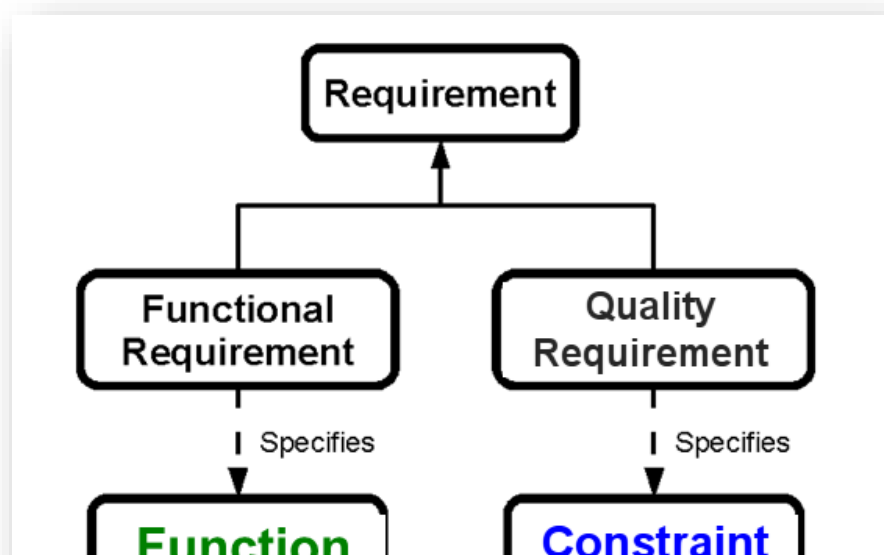
- › *What does the quality attribute scenario add to the ASR summary description?*

## **2. More explicit discussion of the link to Functionality**

# Some common questions

- › *What does the quality attribute scenario add to the ASR summary description?*

## 2. More explicit discussion of the link to Functionality



# Some common questions

## 2. More explicit link to Functionality

### Modifiability: New types of documents

Response:

- › This modification should not affect the interface for initializing and initiating document processing jobs, except for the functionality related to checking the raw data for errors.
- › The interface for registering companies should be extended with the functionality to enable the new types of documents for new customer organizations and configure their template for these types of documents and check these templates for errors.
- › The interface for consulting the personal document store and specific documents should be extended with the functionality to show the new types of documents.
- › This modification should not affect the systems that generate documents: generating the new types of documents should reuse as much of the existing functionality for generating documents as possible.
- › Storing generated documents of the new types of documents should not require changes to the existing system for storing generated documents.

# Some common questions

- › *What does the quality attribute scenario add to the ASR summary description?*
- 3. **More fine-grained characterisation of the NFR**, in terms of different responses, different sub-cases..

# Some common questions

## Modifiability: New types of documents

### 3. More fine-grained characterisation

---

DocProc wants to offer a new document type without impacting the existing document generation infrastructure. Changes should have **no** impact on the existing document generation infrastructure, but only affect the external elements (templates, raw data, etc.). Extending document processing job initialization should take less than **1** person month to implement, test, and deploy.

---

#### Response:

- › This modification should not affect the interface for initializing and initiating document processing jobs, except for the functionality related to checking the raw data for errors.
- › The interface for registering companies should be extended with the functionality to enable the new types of documents for new customer organizations and configure their template for these types of documents and check these templates for errors.
- › The interface for consulting the personal document store and specific documents should be extended with the functionality to show the new types of documents.
- › This modification should not affect the systems that generate documents: generating the new types of documents should reuse as much of the existing functionality for generating documents as possible.
- › Storing generated documents of the new types of documents should not require changes to the existing system for storing generated documents.

# Some common questions

- › *What does the quality attribute scenario add to the ASR summary description?*

**4. Response measures!** Measurable, objectively verifiable – *how can we objectively evaluate whether the requirement is met?*



# Some common questions

## 4. Measurable, objectively verifiable

### Response:

- › This modification should not affect the interface for initializing and initiating document processing jobs, except for the functionality related to checking the raw data for errors.
- › The interface for registering companies should be extended with the functionality to enable the new types of documents for new customer organizations and configure their template for these types of documents and check these templates for errors.
- › The interface for consulting the personal document store and specific documents should be extended with the functionality to show the new types of documents.
- › This modification should not affect the systems that generate documents: generating the new types of documents should reuse as much of the existing functionality for generating documents as possible.
- › Storing generated documents of the new types of documents should not require changes to the existing

### Response Measure:

- Extending the interface for initializing document processing jobs takes **less than 1 man month** to implement, test and deploy.
- Changing the system to manage templates to support new types of document should take **less than 2 man months** to implement, test and deploy.
- Extending the interface for viewing the personal document store and a specific received document takes **less than 4 man months** to implement, test and deploy.

*Keep going...*